EX. NO: 8 — IMPLEMENTATION OF DIFFIE-HELLMAN KEY EXCHANGE

Aim:

To implement the Diffie-Hellman Key Exchange algorithm using C.

Algorithm:

1. Enter the public values n and g.
2. Enter Alice's private key x.
3. Calculate Alice's public key: A = g^x mod n.
4. Enter Bob's private key y.
5. Calculate Bob's public key: B = g^y mod n.
6. Calculate the shared secret key.
7. Display both keys.

Program:

```
#include <stdio.h>
long long power(long long a, long long b, long long n)
{
    long long result = 1;
    while(b > 0)
    {
        result = (result * a) % n;
        b--;
    }
    return result;
}
int main()
{
    long long n, g, x, y;
    long long A, B, key1, key2;
    printf("Enter the value of n and g: ");
```

```
    scanf("%lld %lld", &n, &g);
    printf("Enter private key of first person: ");
    scanf("%lld", &x);
    A = power(g, x, n);
    printf("Enter private key of second person: ");
    scanf("%lld", &y);
    B = power(g, y, n);
    key1 = power(B, x, n);
    key2 = power(A, y, n);
    printf("\nPublic key of first person: %lld", A);
    printf("\nPublic key of second person: %lld", B);
    printf("\nShared key of first person: %lld", key1);
    printf("\nShared key of second person: %lld", key2);
    return 0;
}
```

Output:

```
C:\Users\Priyanga\OneDrive\I
Enter the value of n and g: 23 5
Enter private key of first person: 6
Enter private key of second person: 15

Public key of first person: 8
Public key of second person: 19
Shared key of first person: 2
Shared key of second person: 2
--------------------------------
Process exited after 14.55 seconds with return value 0
Press any key to continue . . .
```

Result:

Thus, the Diffie-Hellman Key Exchange algorithm was implemented successfully using C.